

The HIL Index

A Non-Saturating, Harness-Standardized Capability Index
for AI Models and Agents

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Status. This paper is a proposed companion to *Unified Intelligence Theory and the Harness Intelligence Level* and *Unified Agent Benchmark (HIL-Bench)*. It introduces a public model-comparison index without redefining the semantic meanings of the HIL capability levels.

Abstract

AI model comparison is increasingly dominated by composite intelligence indices that aggregate performance across reasoning, mathematics, coding, scientific knowledge, long-context tasks, and agentic work. Such indices are useful because they compress many evaluations into one number, but they face recurring problems: individual benchmarks saturate, benchmark mixtures change over time, static public tasks become optimization targets, historical scores may be hard to interpret, and raw model performance can be conflated with the capabilities of the harness through which the model is deployed.

This paper proposes the **Harness Intelligence Level Index (HIL Index)**, a public model-comparison index derived from the Unified Intelligence Theory and HIL-Bench measurement framework. The HIL Index is designed to complement rather than redefine the invariant HIL capability hierarchy. For singleton systems, the proposed headline index is derived from latent achievements in Cognitive Intelligence C , Individual Intelligence I , Delegation Intelligence DI , and operational Self-Awareness SA , while Memory Capability M remains separately reported to avoid double-counting. The categorical HIL level and full capability profile remain scientifically primary; the HIL Index exists for ranking, trend analysis, communication, and public model comparison.

The proposed architecture combines: (1) invariant semantic capability levels, (2) difficulty-calibrated latent scores without a fixed maximum, (3) permanent anchor forms plus fresh procedurally generated hidden witnesses, (4) a frozen HIL Reference Harness family, and (5) independent validation against downstream real-world outcomes. The intended public result is therefore not merely a number but a structured report such as

154 | $UI3$ | [$C5, I3, T4/H1, SA3$] | $M4$

with uncertainty, cost, latency, false-completion rate, resource envelope, and Harness Gain reported separately. The central hypothesis is that a standardized agent-realization index can measure a capability object not captured by either a weighted collection of contemporary benchmarks or a general latent benchmark scale: *how much verified intelligence a model can realize when instantiated as a controlled AI agent, and how much work can consequently be delegated to it with limited human cognitive intervention.*

Keywords: Harness Intelligence Level; HIL Index; AI benchmark; latent capability; agent benchmark; model evaluation; benchmark saturation; delegation; intelligence measurement.

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1 Introduction

AI capability measurement faces a tension between simplicity and scientific fidelity. Model users, researchers, journalists, and companies benefit from a single number that makes frontier models easy to compare. Yet the capabilities increasingly relevant to deployed AI systems are not one-dimensional. Solving a difficult mathematics problem is different from maintaining useful state across a restart. Persistence is different from learning from verified experience. Learning is different from improving the mechanism responsible for cognition. Strong one-shot reasoning is different from completing a long task with little human cognitive intervention.

The Unified Intelligence Theory addresses this problem by separating several capability families rather than collapsing them at the semantic level. Its measured core is

$$[C, I, O, T, H, SA],$$

where Delegation Intelligence is represented through the joint Task Difficulty T and Human Cognitive Intervention H surface. For singleton systems, the existing HLIS construction uses the conceptual dimensions

$$D = \{C, I, DI, SA\},$$

while organizational systems additionally include O . Memory Capability M is independently measured and supports higher Individual Intelligence but is not itself an additional Unified Intelligence coordinate [1, 2].

The existing framework also distinguishes a frozen base model m from a harness h :

$$A = A(m, h).$$

A behavioral capability is generally a property of this pair. Attributing the resulting intelligence solely to the base model without controlling the harness can therefore confound model capability with persistence, tools, verification, retry procedures, memory, or other runtime infrastructure.

This paper asks a narrower practical question:

Can the HIL framework produce one public index that is as simple to use as existing intelligence leaderboards while remaining non-saturating, harness-controlled, interpretable, and predictive of real agent performance?

We propose that it can, but only if the public HIL Index is treated as a distinct measurement layer rather than as a replacement for the underlying scientific profile.



Figure 1: Proposed division of labor. The HIL Index is a measurement and communication layer built on top of the theory and benchmark, not a redefinition of the underlying capability semantics.

1.1 Contributions

This paper proposes:

1. a non-saturating latent HIL Index with no fixed maximum;
2. a fixed HIL Reference Harness family for fair base-model comparison;
3. anchored coordinate scales for C , I , DI , and SA ;
4. a separation between semantic capability level and difficulty within the level;
5. a three-surface evidence model: public references, permanent private anchors, and fresh hidden witnesses;

6. a public Delegation Frontier and Harness Gain alongside the headline index;
7. a validation tournament against external real-world outcomes and competing indices;
8. versioning and governance rules intended to preserve historical interpretability.

2 Relationship to Existing Capability Indices

2.1 Artificial Analysis Intelligence Index

As of September 2026, Artificial Analysis Intelligence Index v4.2 combines ten evaluations across four weighted categories: Agents (30%), Coding (20%), Scientific Reasoning (20%), and General (30%) [3]. The index is independently run, reports uncertainty, and emphasizes practical current-model comparison. Artificial Analysis also reports cost and speed separately from intelligence, a useful product-design principle that HIL should preserve.

Version 4.2 illustrates the maintenance challenge of contemporary benchmark mixtures. Artificial Analysis removed GPQA Diamond after describing it as saturated, added AA-Briefcase and GDP.pdf, and doubled private held-out weighting to 40% in order to increase realism and reduce gaming [4]. These are reasonable improvements, but they also show that frontier discrimination requires recurring changes to the component suite.

The HIL Index proposes a different response:

semantic capability remains fixed while witness difficulty can increase.

2.2 Epoch Capabilities Index

The Epoch Capabilities Index (ECI) addresses saturation more directly. Current Epoch documentation describes ECI as a general capability scale built from scores spanning roughly 60 benchmarks, with a statistical model inspired by Item Response Theory [5, 7]. Its core relationship is

$$\text{performance}(m, b) = \sigma(\alpha_b[C_m - D_b]),$$

where C_m is model capability, D_b is benchmark difficulty, and α_b is a slope parameter [6]. ECI reports bootstrap uncertainty and is explicitly unbounded; Epoch currently rescales the latent fit using fixed model calibration points for convenient presentation.

ECI is therefore a strong response to saturation. HIL should not attempt to compete with it by simply aggregating more benchmarks. The intended HIL measurement object is different: **standardized agent-realizable intelligence**.

2.3 Different questions

The three approaches can be summarized as:

AAI : How strong is this model on a curated current frontier suite?

ECI : Where is this model on a cross-benchmark latent capability scale?

HIL Index : How much verified intelligence can this model realize as a standardized agent?

Table 1: Comparison of measurement philosophies.

Property	Artificial Analysis Index	Epoch Capabilities Index	Proposed HIL Index
Primary object	Current composite benchmark performance	General latent benchmark capability	Standardized agent-realizable intelligence
Headline score	Yes	Yes	Yes
Fixed maximum	100	No	No
Difficulty calibration	Mainly suite-level/versioned	Core statistical method	Core method within invariant capability levels
Semantic capability hierarchy	No explicit HIL-like hierarchy	No explicit hierarchy	Yes
Standardized persistent harness	Partial/task dependent	Not primary	Core requirement
Persistent Individual Intelligence	Partial agent evidence	Not explicit	Core coordinate
Delegation T/H	Agent tasks	Not explicit	Core surface
Self-awareness / false completion	Partial	Not explicit	Core coordinate / metric
Independent memory ladder	No	No	Yes, separately reported
Harness Gain	No	No	Yes
Fresh hidden generation	Increasing private use	Mixed benchmark sources	Core design goal

3 Measurement Target of the HIL Index

We define the HIL Index as:

A standardized, non-saturating estimate of how much verified intelligence a frozen AI model can realize when evaluated through the HIL reference harness family.

For singleton models, define the latent capability vector

$$\Theta_m = [\theta_C, \theta_I, \theta_{DI}, \theta_{SA}].$$

The dimensions measure:

- θ_C : cognitive problem solving, abstraction, strategy, discovery, and cross-domain integration;
 - θ_I : persistent continuity, adaptation, governed improvement, discovery, and higher longitudinal functions where demonstrated;
 - θ_{DI} : verified task difficulty under bounded human cognitive intervention;
 - θ_{SA} : grounded self-modeling, calibration, failure recognition, and related operational self-awareness.
- Memory remains separately reported:

$$M(m) \in \{M0, M1, M2, M3, M4, M5, M\Omega\}.$$

For organizational systems, a later extension may include θ_O .

4 Three Measurement Objects Must Remain Distinct

4.1 HIL-Level

HIL-Level is an ordinal, semantically invariant capability claim. Examples include $UI3$, $C5$, $I3$, $T4/H1$, and $SA3$. These claims require canonical evidence and lower-level retention.

4.2 HLIS

HLIS measures continuous achievement for one frozen model-harness pair. The existing theory uses a geometric aggregation

$$HLIS(m, h; D, \alpha, R, B) = 100 \left(\prod_{d \in D} A_d^{\alpha_d} \right)^{1/\sum_d \alpha_d},$$

where $A_d \in [0, 1]$ is cumulative achievement on dimension d . This provides partial bottleneck sensitivity but does not replace the hard Unified gate [1].

4.3 HIL Index

The HIL Index serves a different purpose:

HIL Index = public, model-level, statistically calibrated comparison scale.

Thus:

HIL-Level : What capability has been demonstrated?

HLIS : How strong is this model-harness pair?

HIL Index : Where does this base model sit on a common non-saturating scale?

5 HIL Reference Harness Family

A fair base-model comparison requires a standardized runtime. We define a HIL Reference Harness family

$$\mathcal{H}_1^{ref} = \{HG0, HG1, \dots, HG6, HG\Omega\}.$$

The existing HIL framework already describes a ladder in which progressively richer mechanisms enable persistent state, consolidation, governed modification, meta-improvement, discovery, organization, and frontier expansion, while explicitly stating that architecture enables a claim but external evidence earns it [1].

For every evaluated model, the reference harness must standardize, where applicable:

- context budget and model-call budget;
- code execution and file interfaces;
- web/search permissions;
- persistent-state capacity and retrieval interface;
- verification and retry interface;
- time and compute ceilings;
- lifecycle restart mechanism;
- human intervention allowance;
- authority and write permissions.

A HIL Index result is therefore indexed by the reference harness version:

$$HILIndex_{1.0}(m; \mathcal{H}_1^{ref}).$$

A construct-relevant harness change requires a new index version or demonstrated measurement equivalence.

6 Latent Capability Estimation

A bounded percentage loses discrimination near its ceiling. HIL Index therefore uses an unbounded latent scale.

For binary canonical evidence unit i associated with coordinate d , begin with

$$P(Y_{mdi} = 1) = \sigma[a_i(\theta_{md} - b_i)],$$

where θ_{md} is model capability, b_i is calibrated difficulty, and a_i is discrimination. For richer outcomes, graded response, partial credit, binomial, or hierarchical models may be substituted without changing the conceptual architecture.

The atomic unit need not be a single question. For longitudinal coordinates it may be a complete campaign, for example

$$I2 : \quad \text{baseline} \rightarrow \text{verified experience} \rightarrow \text{restart} \rightarrow \text{held-out transfer},$$

or

$$I3 : \quad \Theta_0 \rightarrow \Theta_1 \rightarrow \text{independent sealed evaluation}.$$

The calibrated observation is therefore a *canonical evidence unit*.

7 Preventing Level Skipping

A high score on upper-level witnesses cannot compensate for failure of lower required capabilities. HIL-Bench already formalizes lower-level retention through

$$A \models_F d_n \iff V_{d,n} = 1 \wedge K_{d,n} = 1[2].$$

For continuous coordinate evidence, define a cumulative achievement transformation such as

$$q_{d,k}^* = \min_{j \leq k} q_{d,j},$$

or a probabilistic equivalent. The latent estimator may use strength within passed constructs, but categorical certification remains constrained by the retention gate.

This preserves the distinction between

task difficulty and capability type.

8 Headline HIL Index

After the coordinate latent variables have been anchored to comparable scales, define

$$\theta_H(m) = \sum_{d \in \{C, I, DI, SA\}} w_d \theta_{md}, \quad \sum_d w_d = 1.$$

For HIL Index v1, begin with equal weights:

$$w_C = w_I = w_{DI} = w_{SA} = 0.25.$$

These weights are a transparent measurement convention, not a natural law.

Map the global latent score to a human-readable scale:

$$HILIndex(m) = 100 + s\theta_H(m).$$

A provisional presentation choice is $s = 10$. Thus $\theta_H = 0$ maps to 100 and $\theta_H = 1$ maps to 110. There is no maximum:

$$HILIndex \in (-\infty, \infty).$$

The origin and scaling constant must be empirically calibrated before v1 ratification.

9 Permanent Anchor Calibration

Historical stability requires scale anchors that do not move whenever the model population changes. For coordinate d , let the permanent anchor set be $\mathcal{A}_d$. After ratification, fix

$$\{a_i, b_i : i \in \mathcal{A}_d\} = \text{constant}.$$

New witness forms are calibrated against anchors and reference systems while anchor parameters remain fixed.

The HIL-Bench design already distinguishes permanent public references, private anchors, and new construct-equivalent witnesses, and requires form equating for continuous scores [2]. The HIL Index should make this the foundation of its statistical scale.

A useful two-track policy is:

1. **Frozen historical HIL Index:** never recomputed after publication under the same index version.
2. **Current-fit diagnostic:** optional research estimate using all contemporary data, clearly separated from the frozen historical score.

10 Solving Benchmark Saturation

The HIL Index should separate capability level from difficulty within the capability. Suppose

$$C3 = \text{strategic reasoning}.$$

Define a stress vector

$$d_{C3} = (N_{states}, d_{dependency}, b_{branch}, \nu_{novelty}, r_{distractor}).$$

A model may solve $C3@d = 4$ today and another solve $C3@d = 18$ later. Both are still C3. C4 remains a different construct, not merely “harder C3.”

The same rule applies throughout HIL:

$\text{anchor saturation} \neq \text{construct obsolescence}.$

A saturated witness remains a historical anchor. Frontier discrimination moves to harder construct-equivalent forms or higher capability levels. HIL-Bench explicitly calls for headroom while retaining saturated forms as anchors [2].

11 Procedurally Generated Hidden Witnesses

Static benchmarks become optimization targets. The strongest HIL Index forms should use procedural generation whenever the construct permits it:

$$x_r \sim G_{d,k}(s_r, \lambda_r),$$

where s_r is a hidden seed and λ_r controls calibrated difficulty.

For C5 discovery, generate a new synthetic world

$$\omega_r \sim G_{C5}$$

with latent law F_{ω_r} . The system observes evidence, derives a method/model, and predicts sealed cases. The exact world did not exist before the run.

Similar generators can create:

- novel causal/planning structures for C3;

- cross-domain dependency worlds for C6;
- new provenance and supersession histories for M2;
- new experience-to-transfer laws for I2;
- controlled uncertainty/failure interventions for SA.

Procedural generation supplements rather than replaces permanent anchors.

12 Three Evidence Surfaces

Every important construct should draw from:

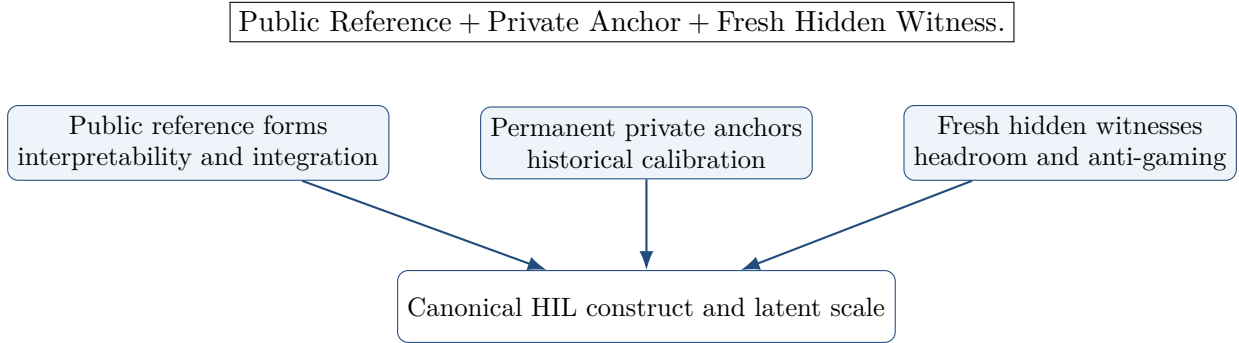


Figure 2: Three evidence surfaces supporting one invariant construct.

Public forms teach the interface and construct. Private anchors preserve continuity. Fresh witnesses preserve headroom and contamination resistance.

13 Cognitive Component

The cognitive component estimates θ_C using the existing HIL family panel: general, science, mathematics, abstract reasoning, code, multimodal/spatial reasoning, and long-context reasoning [2]. Strength in one family should not automatically substitute for another.

The C ladder provides qualitative structure:

$$C0 \rightarrow C1 \rightarrow C2 \rightarrow C3 \rightarrow C4 \rightarrow C5 \rightarrow C6 \rightarrow C\Omega.$$

Difficulty calibration within each level supplies long-term discrimination.

14 Individual Intelligence Component

The strongest conceptual differentiator of HIL is likely θ_I . Rather than restricting intelligence to static problem solving, HIL measures one persistent individual progressing through

$$I0 \rightarrow I1 \rightarrow I2 \rightarrow I3 \rightarrow I4 \rightarrow I5 \rightarrow I\Omega.$$

This includes, where demonstrated, persistence across lifecycle discontinuities, adaptation from verified experience, governed cognitive-mechanism improvement, improvement of the improvement process, persistent autonomous discovery, and repeated frontier expansion.

These are controlled longitudinal experiments, not ordinary quiz items. HIL therefore asks a question that static benchmark aggregates do not naturally answer:

Can this model merely answer difficult questions, or can it function as an improving persistent agent?

15 Delegation Component and Delegation Frontier

Delegation Intelligence should not be reduced to a generic agent accuracy score. HIL measures a surface over (T, H, p) , where T is task difficulty, H is human cognitive intervention, and p is verified reliability.

Define the public Delegation Frontier

$$DF_{h,p}(A) = \max\{T : P(\text{delivered success} \mid A, T, H \leq h) \geq p\}.$$

A model card may therefore report

$$T4/H1@p = 0.80.$$

This has direct operational meaning: the hardest task class the model can reliably complete while requiring at most the declared human cognitive assistance.

The latent θ_{DI} summarizes the broader surface; the frontier remains visible because it is easier to interpret than the scalar alone.

16 Self-Awareness and False Completion

The HIL Index should distinguish an underlying correct attempt from a trustworthy delivered outcome. The prior HIL pilot found that two harness conditions could show similar raw success while differing in false-completion behavior, motivating delivered-outcome accounting [1].

The SA component should therefore include externally grounded tests of calibration, recognition of missing information, failure detection, appropriate retry/withhold behavior, and causal self-diagnosis where the level requires it.

Define False Completion Rate

$$FCR = \frac{N_{\text{incorrect delivered as complete}}}{N_{\text{attempted tasks}}}.$$

FCR should be reported prominently rather than hidden inside the headline score.

HIL measures what the AI can be trusted to deliver, not merely what it generates.

17 Memory Reporting

Memory should remain outside the HIL Index headline aggregation. Report

$$M0, M1, M2, M3, M4, M5, M\Omega$$

separately. This prevents memory capability from being counted once explicitly and again indirectly through Individual Intelligence. The current HIL architecture already treats M as a supporting prerequisite rather than a sixth top-level intelligence family [1, 2].

18 Harness Gain

A uniquely useful HIL metric is

$$HarnessGain(m) = \max_k HLIS(m, HG_k) - HLIS(m, HG_0).$$

Harness Gain estimates how much capability a model can realize when supplied with standardized persistence, verification, tools, learning infrastructure, and other harness mechanisms.

Two models may have similar raw cognition but very different Harness Gain. For agent builders, that difference can be practically important. Harness Gain should therefore appear beside HIL Index but should not be merged into it.

19 Resource Reporting

Every HIL Index run should record tokens, calls, wall time, compute class, persistent-state capacity, tool/knowledge access, human cognitive load, and subagent count when relevant. A capability number without its resource envelope is incompletely specified.

However,

$$\boxed{\text{capability} \neq \text{cost} \neq \text{speed}.}$$

Report separate Pareto analyses such as HIL Index vs. cost, HIL Index vs. latency, and HIL Index vs. token use. This follows the useful product principle that intelligence and efficiency should be visible together but not conflated.

20 Evaluation Tiers and Publication Speed

HIL-Bench already defines HIL-Quick, HIL-Core, HIL-Cert, HIL-Extended, and HIL-Omega as evidence/cost packages rather than intelligence levels [2]. The HIL Index should reuse these packages.

A public leaderboard entry may have status:

- **Preliminary:** HIL-Core evidence, explicitly limited to tested capabilities;
- **Certified:** HIL-Cert plus Extended/Omega evidence where required.

A preliminary score must never infer untested longitudinal capabilities. Fast publication is important for adoption, but it cannot come at the cost of pretending that expensive causal evidence was collected when it was not.

21 Uncertainty

Every HIL Index score must include uncertainty, for example

$$HILIndex = 154.2 [151.8, 156.6].$$

For ordinal level certification, use the preregistered HIL reliability rule. For latent-score uncertainty, bootstrap, Bayesian posterior, or another validated procedure may be used.

Leaderboard UI should avoid implying meaningful ordering when confidence intervals substantially overlap.

22 Validation Program

The measuring instrument must itself be testable. HIL-Bench already requires specification-key consistency, test-retest reliability, rank/profile stability, construct-selective ablations, known controls, human calibration, held-out-family generalization, predictive validity, and headroom [2].

The HIL Index adds index-specific requirements:

22.1 Test-retest reliability

Independent hidden runs should yield compatible coordinate and global estimates.

22.2 Alternate-form reliability

Construct-equivalent generated forms should produce statistically compatible rankings and profiles.

22.3 Provider stability

Where the same frozen model is served by multiple providers, results should be stable after controlling for serving differences.

22.4 Construct validity

Ablating a mechanism relevant to one coordinate should selectively impair that coordinate more than unrelated coordinates.

22.5 Rank robustness

Seed-set changes should not create large rank reversals without corresponding uncertainty.

23 Predictive-Validity Tournament

The strongest evidence for HIL is external prediction, not self-declared superiority.

For models $m = 1, \dots, N$, collect

$$H_m = HILIndex(m), \quad E_m = ECI(m), \quad A_m = AAI(m).$$

Create an independent downstream suite $D^{external}$ that is not used to construct any of the three scores. Candidate outcomes include repository-scale software work, research/analysis workflows, document-intensive professional work, long-horizon task completion, error recovery, human intervention required, delivered correctness, and failure-recognition accuracy.

Let downstream performance be Y_m . Compare

$$\rho(H, Y), \quad \rho(E, Y), \quad \rho(A, Y),$$

and predictive quantities such as

$$R_H^2, \quad R_E^2, \quad R_A^2.$$

The central scientific test is

Does HIL predict important external outcomes better?

Only repeated independent evidence of this kind would justify claims that HIL is a more useful model-selection instrument for deployed agents.

24 Public Leaderboard Design

The first public view should remain simple.

Table 2: Illustrative HIL Index leaderboard schema. Values are examples only.

Rank	Model	HIL Index	HIL Level	C	I	Delegation	SA	M	Cost
1	Model A	158	UI3	C5	I3	T4/H1	SA4	M4	–
2	Model B	154	UI2	C6	I2	T4/H2	SA3	M3	–
3	Model C	149	UI3	C5	I3	T3/H1	SA3	M4	–

Each model page should additionally display:

- confidence interval;
- $\theta_C, \theta_I, \theta_{DI}, \theta_{SA}$;
- Harness Scaling Curve;
- Harness Gain;
- False Completion Rate;
- verification responsiveness;
- cost, latency, and token consumption;
- benchmark/index version and resource envelope.

The first view remains one number; the scientific structure remains underneath.

25 Compact Reporting String

Every official result should support

$$HILIndex \mid HILLevel \mid [C, I, T/H, SA] \mid M.$$

For example

$$154 \mid UI3 \mid [C5, I3, T4/H1, SA3] \mid M4.$$

This combines relative ranking, absolute capability semantics, multidimensional profile, and memory diagnostic. A standalone HIL Index value should never be treated as the complete scientific result.

26 Versioning and Historical Stability

Each score must identify the HIL semantic standard, HIL Index version, Reference Harness version, and witness generation, for example

$$HIL-1.0/Index-1.0/HRH-1/2026Q4.$$

Semantic revisions require a new major HIL standard. Changes to index calibration or aggregation require a new Index version. Routine renewal of hidden witnesses does not redefine the semantic level.

Historical published scores should remain archived rather than silently overwritten.

27 Governance

A credible public index requires institutional separation. Official evaluation should distinguish:

1. the party requesting or funding evaluation;
2. the runner executing the model/harness;
3. the owner of hidden witness forms;
4. the verifier or judging process;
5. the certification authority.

These separations are already recommended by HIL-Bench [2].

As adoption grows, a HIL Standards Council should include external researchers and independent evaluators. Payment, sponsorship, or provider participation must not affect task selection, seeds, thresholds, judging rubrics, promotion criteria, or index calibration.

28 Publication Speed, Coverage, and Product Requirements

A technically excellent index can remain irrelevant if it evaluates too few models or reports results too slowly. Operational requirements therefore include:

1. rapid preliminary evaluation of major releases;
2. broad provider and open-weight coverage;
3. reproducible API/model-version identifiers;
4. public historical time series;
5. transparent handling of inaccessible or retired models;
6. a simple web interface and machine-readable API.

A useful public API could expose endpoints for models, index history, coordinate scores, delegation frontiers, memory levels, and benchmark metadata.

29 Falsifiable Predictions

The HIL Index makes the following testable predictions:

1. **Harness realization:** models with similar raw cognition will differ materially in Harness Gain.
2. **Agent prediction:** $\theta_I + \theta_{DI}$ will predict long-horizon agent performance better than cognition-only scores.
3. **Human burden:** Delegation Frontier will predict required human cognitive intervention better than generic agent accuracy.
4. **Delivered reliability:** false-completion-aware metrics will explain deployment reliability beyond raw success.
5. **Saturation resistance:** dynamic construct-equivalent witnesses will retain discrimination without semantic redefinition of HIL levels.
6. **External validity:** the complete HIL Index will predict selected real-world agent outcomes at least as well as competing contemporary indices.

If these predictions systematically fail, the index should be revised before ratification.

30 Limitations

No scalar is a natural law of intelligence. HIL Index weights and scale are measurement conventions. Coordinate equating must be validated before aggregation is defensible. Upper Individual Intelligence levels require expensive longitudinal evidence. Procedural generators can develop artifacts and require auditing. A standardized harness cannot perfectly represent every production agent architecture. Human-reviewed high-level tasks introduce evaluator variance. Highly specialized systems may be poorly summarized by the general singleton index. Governance and incentive problems remain even with strong methodology.

Most importantly, superior conceptual structure does not demonstrate superior practical validity. The index must earn that claim empirically through reliability, construct validity, contamination resistance, headroom, reproducibility, and downstream predictive validity.

31 Development Roadmap

We recommend the following sequence:

1. freeze the Index v1 measurement specification;
2. complete C-Bench difficulty calibration and generators;
3. complete longitudinal I/M infrastructure;
4. implement systematic $T \times H \times p$ delegation measurement;
5. build SA and false-completion suites;
6. establish permanent anchors;
7. evaluate a broad weak-to-frontier reference model population;
8. fit and validate latent coordinate scales;
9. test alternate-form and test-retest reliability;
10. release a public beta leaderboard;
11. run the external predictive-validity tournament;
12. commission independent audit;
13. ratify HIL Index v1 only after the above evidence is satisfactory.

This follows the broader HIL-Bench principle that semantics, canonical laws, controls, audits, repeatability, and scoring should be tested before a standard is frozen.

32 Conclusion

The HIL Index is proposed as a public-facing measurement layer built on top of Unified Intelligence Theory and HIL-Bench. Its purpose is not to replace the multidimensional framework with a scalar. Rather, it provides a scalar where a scalar is useful—ranking models, tracking progress, communicating results, and supporting public leaderboards—while preserving structured capability evidence underneath.

The central result format is

$$\boxed{HILIndex \mid HILLevel \mid [C, I, T/H, SA] \mid M.}$$

The HIL Index differs from a conventional weighted benchmark average by separating stable capability semantics from task difficulty, and from a generic latent benchmark index by explicitly measuring standardized agent realization: cognition, persistent individual intelligence, delegation, and grounded self-awareness.

The proposed index is non-saturating, its semantic levels remain invariant, its hidden witnesses may increase in difficulty without changing those semantics, its base-model comparison is controlled through a standardized reference harness family, and its ultimate usefulness is determined by external prediction:

$$\boxed{\text{Does HIL better predict what AI systems can actually be trusted to do?}}$$

If HIL demonstrates high reliability, contamination resistance, stable historical interpretation, rapid frontier-model coverage, and stronger predictive validity on independent real-world agent tasks, it can provide a useful complement to—and potentially become more widely used than—existing composite intelligence indices. Such a conclusion should be earned by evidence rather than asserted in advance.

A Provisional HIL Index v1 Specification

Field	Provisional specification
Subject	Frozen base model identified by exact model/version/provider configuration.
Reference harness	$\mathcal{H}_1^{ref} = \{HG0, \dots, HG6, HG\Omega\}$ with declared resources and authority.
Primary latent coordinates	$\theta_C, \theta_I, \theta_{DI}, \theta_{SA}$.
Memory	Separately reported $M0$ – $M\Omega$; not added directly to headline score.
Aggregation	$\theta_H = \sum_d w_d \theta_d$; development default equal weights.
Public transform	$HILIndex = 100 + s\theta_H$; development default $s = 10$; no maximum.
Semantic profile	HIL-Level plus $[C, I, T/H, SA]$.
Additional metrics	Harness Gain, FCR, verification responsiveness, cost, latency, tokens.
Evidence surfaces	Public references, permanent private anchors, fresh hidden witnesses.
Historical policy	Frozen published score under versioned anchors; no silent rewriting.
Uncertainty	Mandatory interval and rank-uncertainty reporting.

Field	Provisional specification
Governance	Separation of requester, runner, hidden-form owner, verifier, and certification authority.

B Illustrative Machine-Readable Result

```
{
  "hil_standard": "1.0",
  "hil_index_version": "1.0-beta",
  "reference_harness": "HRH-1",
  "model_id": "provider/model-version",
  "hil_index": 154.2,
  "hil_index_ci": [151.8, 156.6],
  "hil_level": "UI3",
  "profile": {
    "C": "C5",
    "I": "I3",
    "delegation": "T4/H1@p=0.80",
    "SA": "SA3",
    "M": "M4"
  },
  "latent": {
    "theta_C": 6.1,
    "theta_I": 4.7,
    "theta_DI": 5.5,
    "theta_SA": 5.4
  },
  "harness_gain": 14.2,
  "false_completion_rate": 0.023,
  "status": "certified",
  "resource_envelope": "HRH-1-default"
}
```

The values above are illustrative and do not represent any real model.

C Recommended Public Navigation

A HIL website should expose at least the following views:

- overall HIL Index leaderboard;
- HIL Index vs. cost;
- HIL Index vs. latency;
- Cognition (θ_C);
- Individual Intelligence (θ_I);
- Delegation Frontiers;
- Self-Awareness and False Completion;
- Memory Capability;
- Harness Gain;
- historical model progress;
- methodology, versions, and confidence intervals.

References

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- [3] Artificial Analysis, “Artificial Analysis Intelligence Benchmarking Methodology - Intelligence Index v4.2,” September 2026. Accessed September 5, 2026. <https://artificialanalysis.ai/methodology/intelligence-benchmarking>
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- [5] Epoch AI, “Epoch Capabilities Index Documentation - Overview,” 2026. Accessed September 5, 2026. <https://epoch.ai/data/eci-documentation>
- [6] Epoch AI, “Epoch Capabilities Index Documentation - Methodology,” 2026. Accessed September 5, 2026. <https://epoch.ai/data/eci-documentation/methodology>
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